13.H1 Types of Transformations

Lesson Introduction

Benchmark	MA.912.F.2.3 Given the graph or table of $f(x)$ and the graph or table of $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x + k)$, state the number and type of transformation and find the value of the real number k.	Learning Target	• I can state the transformation and find the value of the real number k when given the graph or table of $f(x)$ and the graph or table of $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$.
Benchmark Coherence	Building On: MA.8.F.1 Define, evaluate, and compare functions.	Working Toward: MA.912.T.3.3 Solve and graph mathematical and real-world problems that are modeled with trigonometric functions. Interpret key features and determine constraints in terms of the context.	
Essential Question	• How can I find the value of the real number *k* when a function has undergone a transformation?		
Lesson Narrative	In this lesson, students extend their learning of transformations of functions to find the value of *k* when given the graph or table of the function and the graph or table of its transformation. Students will begin by being introduced to describing transformations, including determining the k-value of the transformation. During this lesson, several function types will be used as part of this new learning.		

Component Summary				
	13.H1.1 Warm-Up (~5 min.)	Transformation of a wave *(SE p. 273)*	13.H1.3 Guided Practice (15-20 min.)	Practice activity describing transformations *(SE p. 277)*
	13.H1.2 Guided Instruction (10-15 min.)	Introduction to describing transformations *(SE p. 274)*	13.H1.4 Your Turn! (15-20 min.)	Independent practice describing transformations *(SE p. 277)*
	13.H1.5 Wrap-Up (5 min.)	Learning target self-reflection *(SE p. 278)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• (Optional) Graph paper	• N/A

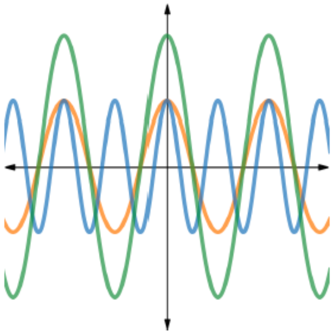
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may misidentify the transformation that is done.
Students may struggle with tables vs graphs when determining the transformation.
Students may apply the wrong sign to the k-value.</td><td>This is an honors lesson that is exclusive for students enrolled in Algebra 1 honors. Consider moving this lesson to post-assessment as a capstone experience.
This lesson helps serve as a review of the function types and key features they have learned throughout this course.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
13.H1.2 Guided Instruction provides students with instruction on describing transformations. There are many instances where students may misunderstand this concept. Leverage this routine to share your thinking out loud. This supports metacognitive and metalinguistic awareness.</td><td>MA.K12.MTR.7.1: Real-World Contexts
This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Students who are struggling readers may find this lesson to be a challenge due to the amount of text and reading comprehension necessary. Consider differentiating the problems using reading and word-problem strategies that are most appropriate for your students and their needs.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.H1.1 Warm-Up: Transformation of a Wave

Component Overview*: Students will review student descriptions of a transformation of a wave to determine with whom they most agree.*

13.H1.1 Warm-Up: Transformation of a Wave

1. The graph shows three different functions. The **orange** graph represents a parent function. The **blue** and **green** graphs are transformations of the parent function.



Terrence and Camila described the graphs.

Terrence	Camila
The green graph is taller than the orange graph, so the y-values were multiplied. The blue graph has more waves than the orange graph and there are more x-intercepts, so the x-values were divided.	The green graph is a stretch of the orange graph vertically, so the y-values were multiplied. The blue graph is a compression horizontally of the orange graph, so the x-values were divided.

Whose description is closest to how you would describe the graphs?

Camila

This activity exhibits the conditions set forth by MA.K12.MTR.4.1. During this activity, students will analyze the mathematical thinking of others and compare the efficiency of a method to those expressed by others.

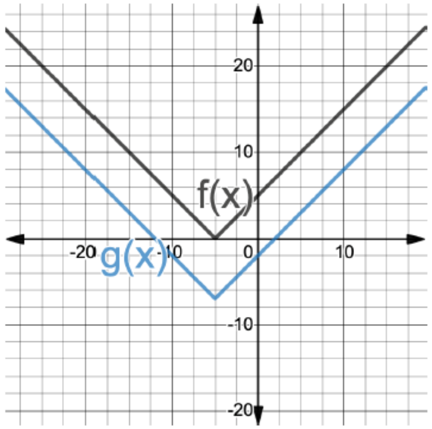
Consider having students share their justification to whose description is closest to how they would describe the graphs.

13.H1.2 Guided Instruction: Describing Transformations

Component Overview: *Students will learn about describing transformations and determining the k-value through five multipart questions.*

13.H1.2 Guided Instruction: Describing Transformations

1. The functions $f(x)$ and $g(x)$ are shown on the graph.



Work with your partner to complete the table to describe $g(x)$.

Type of Transformation	k-value
○ $f(kx)$ ○ $kf(x)$ ○ $f(x+k)$ ○ $f(x)+k$	○ -7 ○ 7

Consider having students do a Notice and Wonder on this graph. This can help students recall what they've learned about absolute value functions and provide information necessary to help launch them into this activity.

13.H1.2 Guided Instruction: Describing Transformations (continued)

2. The table of values for $B(x)$ and $R(x)$ are shown, where $R(x)$ is a transformation of $B(x)$.

$y = B(x)$		$y = R(x)$	
x	y	x	y
-2	-15	-4	-15
-1	-11	-3	-11
0	-7	-2	-7
1	-3	-1	-3
2	1	0	1

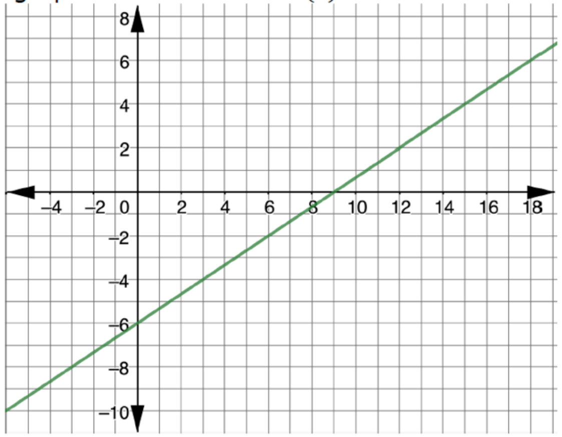
Work with your partner to describe how the graph of $y = B(x)$ was transformed to create $y = R(x)$. Include the k-value in your description.

There was a shift left horizontally, for the x-values changed. The k-value used was a -2. $R(x) = B(x+2)$.

More often, students struggle with functions represented by a table of values more than with a graph. If students understood question 1 but struggle with question 2, use explicit and clear language when teaching question 2, making notes from the table that articulate your thinking for students to reference later.

Consider providing graph paper to students to plot the points for visual learners to see the transformation.

3. The graph of the function $W(x)$ is shown.



a. Write the function, $W(x)$, that represents the line.

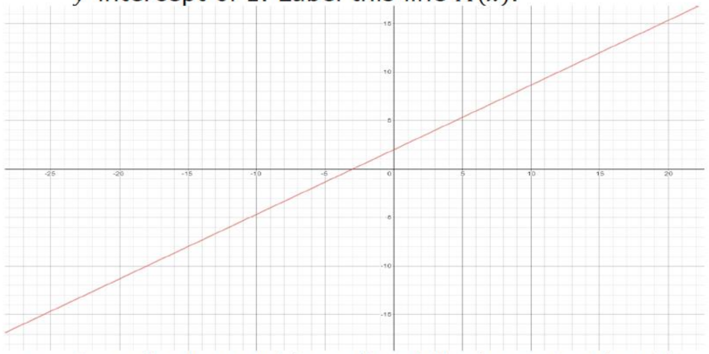
$W(x) = \frac{2}{3}x - 6$

Consider having students do a Notice and Wonder on this graph. This can help students recall what they've learned about linear functions.

Question 3a serves as a scaffold back to writing linear equations from a prior unit.

13.H1.2 Guided Instruction: Describing Transformations (continued)

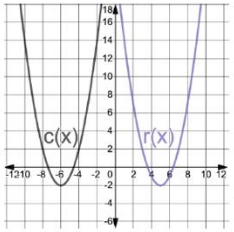
b. Draw a line that is parallel to $W(x)$ with a y-intercept of 2. Label this line $M(x)$.



c. Describe how $W(x)$ can be shifted to coincide with $M(x)$.
Vertical shift up 8 units or a horizontal shift left 12 units.

d. Write the function $M(x)$ in terms of $W(x)$.
$M(x) = W(x) + 8$ *or* $M(x) = W(x + 12)$

4. The functions $c(x)$ and $r(x)$ are shown on the graph.



a. Write each function in vertex form.

$c(x)$	$r(x)$
$c(x) = (x + 6)^2 - 2$	$r(x) = (x - 5)^2 - 2$

b. Describe how $c(x)$ was transformed to create $r(x)$.

Horizontal shift right 11 units. $r(x) = c(x - 11)$

How can both vertical and horizontal shifts be used to describe this transformation?

Consider having students do a Notice and Wonder on this graph. This can help students recall what they've learned about quadratic functions.

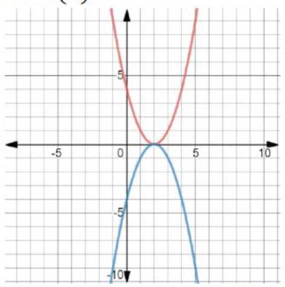
Question 4a serves as a scaffold back to writing a quadratic function in vertex form. Students may need a refresher on vertex form if they have forgotten that from prior units.

13.H1.2 Guided Instruction: Describing Transformations (continued)

c. Discuss with your partner how transformations of quadratic functions can be determined using the vertex form of the functions.
The k-value determines what direction the graph moves vertically up or down. The h-value determines how many spaces the graph moves left or right. If the h-value is a positive value then the graph moves to the left, and If the h-value is a negative the graph moves to the right. If the k-value is a negative sign the graph reflects about the x-axis.

5. The functions $B(x)$ and $H(x)$ are shown on the graph. The table of values for $B(x)$ and $H(x)$ are shown, where $H(x)$ is a transformation of $B(x)$.

$y = B(x)$		$y = H(x)$	
x	y	x	y
-1	9	-1	-9
0	4	0	-4
1	1	1	-1
2	4	2	-4
3	9	0	-9



Work with your partner to complete the table to describe $H(x)$.

Type of Transformation	k-value
○ $B(kx)$ ○ $kB(x)$ ○ $B(x+k)$ ○ $B(x)+k$	-1

What do you think will be true about the k-value just by looking at the graph?
What do you notice about the y-values for the two tables of values?

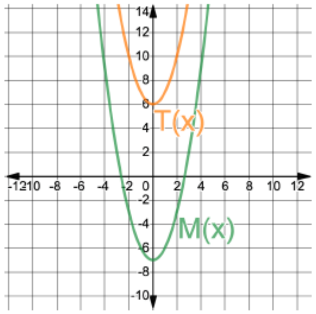
What would the graph look like if the k-value was positive instead of negative?

13.H1.3 Guided Practice: Describing Transformations

Component Overview: *Students will work together to complete two practice questions with describing transformations.*

13.H1.3 Guided Practice: Describing Transformations

1. The functions $M(x)$ and $T(x)$ are shown on the graph, where $M(x)$ is a transformation of $T(x)$.



Write the transformation by completing the equation.

$M(x) = T(x) + (-13)$

2. The table of values for $B(x)$ and $R(x)$ are shown, where $R(x)$ is a transformation of $B(x)$. Write a description of the transformation.

$y = B(x)$		$y = R(x)$	
x	y	x	y
9	10	1	10
7	8	-1	8
5	6	-3	6
3	8	-5	8
1	10	-7	10

Horizontal shift left 8 *units.* $R(x) = B(x + 8)$.

Consider leveraging a Think-Pair-Share routine with this activity. Allow students a couple minutes to try this activity on their own and then, when time is called, work with a partner.

13.H1.4 Your Turn!

***Component Overview**: Students will complete two practice questions, independently, with describing transformations.*

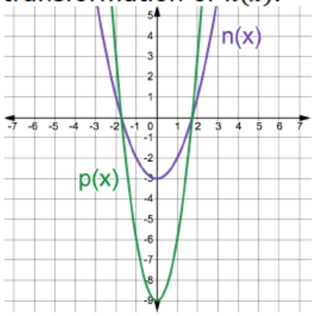
13.H1.4 Your Turn!

1. The table of values for $f(x)$ and $g(x)$ are shown, where $g(x)$ is a transformation of $f(x)$. Write a description of the transformation.

$y = f(x)$		$y = g(x)$	
x	y	x	y
-2	-12	-2	-1
1	-6	1	5
2.5	-3	2.5	8
4	-6	4	5
7	-12	7	-1

Vertical shift up 11 *units.* $g(x) = f(x) + 11.$

2. The functions $n(x)$ and $p(x)$ are shown on the graph, where $p(x)$ is a transformation of $n(x)$.



What is the value of k such that $p(x) = kn(x)$?

3

Consider using the time in this activity to work with a small group of students who performed poorly during the 13.H1.3 Guided Practice activity.

13.H1.5 Wrap-Up: Reflection

***Component Overview**: Students will complete a reflection based on the learning targets for this lesson.*

13.H1.5 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

Answers will vary.

1. I can determine the k-value of a transformation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can describe a transformation using words or function notation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.